# LLM Text Source Classification

Distinguishing Human Text from 11 LLM Architectures

Affan Pathan (CS24BT033) Ayush Yadav (CS24BT001)
Aaditya Kumar (CS24BT012)
Rohit Patil (IS24BM017) Arjun Gangwar (IS24BM014)

Department of Computer Science and Engineering
Indian Institute of Technology Dharwad

*Supervised by: Sujeet Kumar, PhD Scholar*

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# Problem Statement

## Core Challenge

LLMs such as GPT-4, LLaMA, and Mistral produce text that is often **indistinguishable from human writing**.

**Why does this matter?**

- Academic integrity violations
- Spread of AI-generated misinformation
- Content authenticity concerns

## Task Definition

Given a text sample, determine whether it was generated by one of **11 LLMs** or written by a **human** — a 12-class classification problem.

## 12 Classes

1. ChatGPT
2. Cohere
3. Cohere-Chat
4. GPT-2
5. GPT-3
6. GPT-4
7. **Human**
8. LLaMA-Chat
9. Mistral
10. Mistral-Chat
11. MPT
12. MPT-Chat

# Dataset Overview

## Scale & Balance

- **996,000** total text samples
- **12 perfectly balanced** classes
- **83,000** samples per class

| Split | Samples | % |
|---|---|---|
| Train | 796,800 | 80% |
| Validation | 99,600 | 10% |
| Test | 99,600 | 10% |
| **Total** | **996,000** | **100%** |

## Key Linguistic Statistics

- Global mean word count: **212.4** words
- Avg sentence length: **25.97** words/sent
- **GPT-3**: most concise (mean 131)
- **LLaMA-Chat**: most verbose (mean 255)
- **GPT-2 & MPT**: longest sentences ($\sim$36–38 words/sent)
- **GPT-4 & ChatGPT**: shorter, structured sentences ($\sim$18–19 words)

# Dataset: Vocabulary & Lexical Analysis

## Lexical Diversity (5,000 samples)

| Source | Unique Word Types |
|---|---|
| Human | 41,410 |
| MPT | 45,133 (highest LLM) |
| GPT-3 | 25,193 (lowest) |

## TF-IDF Insights

- **GPT-3**: terms like *teaspoon, novel*
- **Chat-models**: structural terms like *overall, including*
- **Human**: narrative markers like *said*; HTML artifact `br br`
- GPT-4 & ChatGPT: generic, broad

## Data Integrity

- Zero null/empty entries
- Intra-split duplicates: 21,191 (train), 359 (val/test)
- Truncation at **512 words** retains $> 99\%$ of samples
- 99th percentile: 463 words; max: 13,261 words

## Length-Distribution Bias

Per-class word count distributions differ significantly, which can inflate frequency-based model accuracy.

# End-to-End Classification Pipeline




| Track 1 | Track 2 | Track 3 |
|---|---|---|
| Classical & Hybrid Feature-Based Methods | Sentence Embeddings + Lightweight Classifiers | Transformer-Based Deep Learning Models |

# Best Performing Pipelines — Summary

| Track | Pipeline | Val / Test Acc |
|---|---|---|
| 1 | Neural Net (TF-IDF, 4096-2048) | 81.19% |
| | LinearSVC (TF-IDF) | 77.80% |
| 2 | E5 + MLP | 63.01% |
| | E5 + SVM | 48.38% |
| 3 | **BiGRU + Attention (Learned Emb.)** | **87.13%** |
| | BiLSTM + Attention (BERT Emb.) | 81.44% |
| | BERT + DNN Fine-tune | 69.56% |

**Key Insight**

BiGRU with additive attention and *learned embeddings* achieves **87.13%** — outperforming all pre-trained transformer pipelines while using no pre-trained representations.

# Track 1: Classical & Hybrid Methods

## TF-IDF & SVM Baseline

- Captures surface-level lexical & n-gram patterns
- `HashingVectorizer` ($2^{18}$ buckets, n-grams 1-2) + sublinear TF
- LinearSVC ($C = 1.0$): Test Acc **77.80%**

## Word2Vec + Stylometric

- Skip-gram W2V (100-dim) mean-pooled + punctuation/length stats
- LightGBM: Test Acc **76.95%**

## TF-IDF + Neural Network

Architecture: Dense(L1) $\rightarrow$ BN $\rightarrow$ Dropout $\rightarrow$ Dense(L2) $\rightarrow$ BN $\rightarrow$ Dropout $\rightarrow$ Dense(12)

| L1 | L2 | Test Acc |
|---|---|---|
| 512 | 256 | 79.47% |
| 1024 | 512 | 80.30% |
| 2048 | 1024 | 80.89% |
| **4096** | **2048** | **81.19%** |

# Track 2: Sentence Embeddings + Lightweight Classifiers

## Sentence Encoders Evaluated

| Model | Dim | Key Property |
|---|---|---|
| MiniLM-L6-v2 | 384 | Distilled, fast |
| MPNet-base-v2 | 768 | Full positional context |
| BGE-base-en | 768 | Contrastive, retrieval |
| E5-base-v2 | 768 | Diverse weak supervision |

## Best Results (Test Accuracy %)

| | MiniLM | MPNet | BGE | E5 |
|---|---|---|---|---|
| Log. Reg | 31.68 | 35.41 | 41.01 | 47.00 |
| SVM | 35.75 | 31.75 | 41.45 | 48.38 |
| Rand. Forest | 55.35 | 56.42 | 56.98 | 57.72 |
| **MLP** | 46.72 | 52.13 | 58.43 | **63.01** |

**Preprocessing:** Embeddings standardised via `StandardScaler`. E5 requires `passage:` prefix.

**MLP Head:** emb_dim $\rightarrow$ 512 $\rightarrow$ 256 $\rightarrow$ 128 $\rightarrow$ 12, AdamW, 100 epochs.

# Track 3: BERT + DNN Fine-Tuning (Architecture)

## Model Architecture

- Base: `bert-base-uncased`
- Last **2 encoder layers** unfrozen for domain adaptation.
- **Attention-mask-weighted mean pooling** used to extract a 768-dim sentence vector.
- DNN Classification Head: $768 \rightarrow 1024 \rightarrow 512 \rightarrow 256 \rightarrow 128 \rightarrow 12$
- Activations: ReLU + Dropout(0.3) between linear layers.
- **Dual learning rate strategy**: A significantly lower learning rate for the BERT layers to prevent catastrophic forgetting, and a higher learning rate for the newly initialized DNN head.

# Track 3: BERT + DNN Fine-Tuning (Training & Results)

## Hyperparameters

| Parameter | Value |
|---|---|
| Epochs | 30 |
| Batch Size | 32 (effective: 128 via Grad Accum) |
| Max Seq. Length | 256 |
| LR (BERT layers) | $2 \times 10^{-5}$ |
| LR (DNN head) | $1 \times 10^{-3}$ |
| Weight Decay | $1 \times 10^{-4}$ |
| LR Scheduler | CosineAnnealingLR |
| Mixed Precision | fp16 |

## Results

Test Acc: **69.56%**
Macro F1: **0.6959**

*Note:* Best epoch was 24. The model was still showing slight improvements at termination, suggesting further training could yield higher accuracy.

# Track 3: E5 + LoRA Fine-Tuning (Strategy)

## Low-Rank Adaptation (LoRA)

- Base model: `intfloat/e5-base` (strong performance in Track 2).
- LoRA applied to **Query (Q), Key (K), and Value (V)** projections of every attention block.
- **99.6% parameters frozen**: Only 442,368 out of 109.9M parameters are trainable (0.40%).
- Output: mean-pooled $\rightarrow$ Dropout(0.2) $\rightarrow$ Linear classification layer.
- Enables fine-tuning of a large language model representation on consumer hardware with minimal memory footprint.

# Track 3: E5 + LoRA Fine-Tuning (Training & Results)

## Hyperparameters

| Parameter | Value |
|---|---|
| Epochs | 12 (early stop, patience=4) |
| Batch Size | 48 |
| Max Seq. Length | 256 |
| LoRA Config | Rank ($r$)=8, Alpha=16, Dropout=0.1 |
| Optimizer | AdamW (LR: $2 \times 10^{-4}$) |
| LR Scheduler | Cosine + 10% warmup |
| Mixed Precision | fp16 |

## Result

Test Acc: **66.36%**

*Takeaway:* LoRA on E5 outperforms frozen E5+MLP (63.01%) by over 3%, demonstrating the value of adapting attention weights to the specific task distribution.

# Track 3: BiLSTM with Attention & BERT Embeddings

## Architecture

- Input: **Frozen BERT word embedding layer** (768-dim, vocab 30,522).
- Tokenizer: BERT WordPiece, max **384 tokens** (covers 90th percentile of data lengths).
- Encoder: 2-layer **Bidirectional LSTM** (256 hidden per direction, outputting 512-dim).
- **Custom Additive Attention**: Computes weighted sum over all time steps; padding is masked with $-10^9$ before softmax.
- Head: Lin(512→256) → LayerNorm → ReLU → Drop(0.4) → Lin(256→12).

## Training Details

8 Epochs, Batch 64, AdamW (LR: $3 \times 10^{-4}$), Max Seq: 384, Trained on Full Dataset.

## Result

Val / Test Acc: **81.44%**

# Track 3: BiGRU with Additive Attention (Architecture)

## Model Design (Best Performing)

- Input: **Randomly initialised, fully trainable** `nn.Embedding` (vocab 30,522, dim **256**).
- Tokenizer: BERT WordPiece, max 384 tokens.
- Encoder: 2-layer **Bidirectional GRU** (256 hidden per direction, output 512-dim).
- **Additive Attention Mechanism** (same as BiLSTM, padding masked).
- MLP Classifier Head (same as BiLSTM).

## Why BiGRU outperforms BiLSTM & Transformers here?

1. GRUs suffer less from vanishing gradients on long texts (up to 384 tokens) compared to LSTMs.
2. **Learned embeddings** adapted perfectly to the specific lexical quirks of LLMs, whereas frozen pre-trained embeddings were constrained to general English semantics.

# Track 3: BiGRU with Additive Attention (Training & Results)

## Hyperparameters

| Parameter | Value |
|---|---|
| Epochs | 8 |
| Batch Size | 64 |
| Max Seq. Length | 384 |
| Embedding Dim | 256 (learned) |
| GRU Hidden Dim | 256 (per dir.) |
| Optimizer | AdamW (LR $3 \times 10^{-4}$) |
| LR Scheduler | CosineAnnealingLR |
| Dropout | 0.4 |
| Dataset | Medium |

## Result

Val Acc: **87.13%** (after 8 epochs)

## Efficiency Note

This architecture converged incredibly fast and required significantly less VRAM than BERT fine-tuning, allowing us to process longer sequence lengths (384 vs 256).

# Overall Model Comparison

| Track | Model | Val Acc | Test Acc | Dataset |
|---|---|---|---|---|
| 1 | NN (TF-IDF, 4096-2048) | 81.19% | 81.19% | Full |
| | SVM (TF-IDF) | 77.80% | 77.80% | Full |
| | LightGBM (W2V+Stylo.) | 76.95% | 76.95% | Full |
| | MLP (W2V+Stylo.) | 60.26% | 60.26% | Full |
| 2 | E5 + MLP | 63.01% | 63.01% | Medium |
| | BGE + MLP | 58.43% | 58.43% | Medium |
| 3 | **BiGRU + Attn (Learned Emb.)** | **87.13%** | **87.13%** | Full |
| | BiLSTM + Attn (BERT Emb.) | 81.44% | 81.44% | Full |
| | BERT + DNN Fine-tune | 69.56% | 69.56% | Full |
| | E5 + LoRA | 66.36% | 66.36% | Medium |

**Caution: Track 1 Bias**

TF-IDF accuracy is likely **inflated by length-distribution bias** rather than genuine stylistic discrimination.

**Compute Environment**

Track 1 & RNNs: Kaggle P100/T4
BERT & E5: Colab T4 (16GB VRAM)

# Per-Class Analysis: BERT + DNN Fine-Tune

## Highest F1 Scores

- **GPT-4**: F1 = 0.817 — most distinct signatures
- **LLaMA-Chat**: F1 = 0.769
- **ChatGPT**: F1 = 0.763

## Hardest to Classify

- **Cohere**: F1 = 0.570
- **Cohere-Chat**: F1 = 0.579
- Substantial cross-confusion between the two — same underlying model family.
- Mistral & MPT confused with GPT-2.

## Human Class Anomaly

- Precision: **0.905** (very high)
- Recall: **0.583** (low)
- Model is *conservative* — rarely predicts "human".
- Frequently misclassifies human text as **Cohere**.
- Suggests a *narrow*, high-confidence representation of human writing.

# Conclusion & Findings

## Summary

- **BiGRU + Additive Attention** achieves highest accuracy (**87.13%**), outperforming heavier transformers.
- **TF-IDF** is competitive but accuracy is *partially inflated*.
- **BERT + DNN** offers the most *semantically meaningful* insight but needs more compute.
- **Frozen sentence embeddings** peak at 63% — task-specific fine-tuning is required.

## Limitations

- Differing dataset sizes across tracks due to compute constraints.
- Dataset may contain *prompt-distribution artifacts*.

## Future Work

- Extended training for transformers.
- **Hierarchical classification**: first predict model family, then variant.

# Thank You

**Thank You for Your Attention**